Bloqade Rydberg Blockade Demo

Geometry-Dependent Interaction Modeling

Anchor: Neutral-atom two-site Rydberg blockade simulation using QuEra's Bloqade Analog Python workflow.

Purpose: Demonstrate how atom spacing, Rydberg interaction scale, local emulation, final-shot measurement probabilities, and an interactive performance point connect to a compact blockade metric.

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Abstract. This technical brief describes a compact neutral-atom simulation workflow implemented with QuEra's Bloqade Analog Python interface. The demonstration studies a two-atom Rydberg blockade primitive and connects atom spacing, Rydberg interaction strength, local analog emulation, final-shot measurement probabilities, and double-excitation suppression. The notebook includes an interactive control panel for the spacing window, number of sampled distances, selected performance point, and number of shots. The goal is not to construct a production-level neutral-atom performance model, but to demonstrate the modeling pattern required for architecture-aware quantum benchmarking: define a hardware-relevant control parameter, propagate it through a physically grounded model, extract measurable probability-level outputs, and interpret those outputs as performance-relevant quantities.

Motivation and Scope

Neutral-atom quantum processors provide a natural setting for architecture-aware performance modeling because geometry is both a hardware parameter and an algorithmic resource. Atom positions determine Rydberg interaction strengths; interaction strengths determine blockade behavior; blockade behavior influences the final measurement distribution.

Modeling chain.

atom geometry $\rightarrow$ Rydberg interaction $\rightarrow$ analog quantum dynamics
$\rightarrow$ measurement probabilities $\rightarrow$ performance-relevant metric.

The present demonstration focuses on the simplest nontrivial case: two neutral atoms separated by distance R, driven by a uniform Rabi amplitude and detuning. The key observable is the probability of double Rydberg excitation, P_{rr}.

This two-atom basis clarifies which state combinations are useful for neutral-atom computing. The computationally relevant configurations are those in which the atoms remain in well-defined ground or single-excitation manifolds that can be addressed, entangled, and read out in a controlled way. In contrast, simultaneous excitation of both atoms into the Rydberg state, $|rr\rangle$, is generally not the desired operating point for blockade-based gates. Strong Rydberg interactions are instead used to suppress this double-excitation channel, converting interaction-induced energy shifts into conditional dynamics. For this reason, tracking P_{rr} provides a compact performance diagnostic: low double-excitation probability indicates effective blockade, while leakage into $|rr\rangle$ signals imperfect gate selectivity or a breakdown of the blockade condition.

The notebook uses the current Bloqade Analog import path for analog neutral-atom programming:

```
from bloqade.analog import start.
```

It is organized as a portfolio-style technical artifact with three cells: reusable tools, interactive user-facing controls, and a run/graphics/output cell.

Physical Model

Two-Atom Basis

The two-atom Hilbert space is spanned by

$$\{|gg\rangle\,,\ |gr\rangle\,,\ |rg\rangle\,,\ |rr\rangle\}\,,$$

where $|g\rangle$ denotes the ground or non-Rydberg state and $|r\rangle$ denotes the Rydberg excited state. The central physical mechanism is the interaction-induced shift of $|rr\rangle$.

For a resonant pulse under strong blockade, the drive primarily connects $|gg\rangle$ to the symmetric single-excitation state

$$|W_2\rangle = \frac{1}{\sqrt{2}}\left(|gr\rangle + |rg\rangle\right).$$

Thus, a useful final distribution can have large $P_{gr} + P_{rg}$, small P_{gg}, and suppressed P_{rr}. This is not a static population statement; it is a final-time measurement result for a selected pulse duration.

Rydberg Interaction Scale

For two atoms separated by distance R, the van der Waals Rydberg interaction is modeled as

$$V(R) = \frac{C_6}{R^6}.$$

For the rubidium Rydberg scale used in Bloqade/Aquila-style examples, the coefficient is commonly written as

$$C_6 = 2\pi \times 862690 \quad \text{rad}\,\mu\text{m}^6/\mu\text{s}.$$

The interaction is therefore an angular frequency in rad/μs. For comparison with Rabi-control scales, the notebook reports in MHz:

$$\frac{V(R)}{2\pi} = \frac{C_6}{2\pi R^6}.$$

With the default notebook value $\Omega = 2.0$ rad/μs, the corresponding Rabi frequency is

$$\frac{\Omega}{2\pi} \approx 0.318 \text{ MHz}.$$

A simple order-of-magnitude blockade radius can be estimated from

$$V(R_b) \sim \Omega, \qquad R_b \sim \left(\frac{C_6}{\Omega}\right)^{1/6}.$$

For the values above,

$$R_b \approx 12\ \mu\text{m}.$$

This number should be interpreted as a scale, not as a sharp threshold. In the default simulator, visible P_{rr} leakage can begin around $R \sim 8\ \mu$m, even though $V(R)$ is still larger than Ω. This is a useful quantum-control lesson: blockade is not a classical on/off exclusion rule. It is an off-resonant suppression mechanism, and the suppressed channel can still carry finite probability depending on the ratio V/Ω, the pulse duration, the detuning, and the finite-shot sampling resolution.

Blockade as a crossover.

$$R \text{ small}: \quad V(R) \gg \Omega \quad \Rightarrow \quad P_{rr} \text{ strongly suppressed},$$

$$R \text{ larger}: \quad V(R) \text{ approaches the drive scale} \quad \Rightarrow \quad P_{rr} \text{ leakage becomes visible}.$$

This crossover favors short atom separations for blockade-based neutral-atom control. The architectural implication is that compact layouts are advantageous for strong interactions, but they must still be balanced against addressing, crosstalk, optical access, rearrangement, and array-design constraints.

Driven Two-Level Dynamics

Each atom is driven by a Rabi coupling with amplitude Ω and detuning Δ. In a standard rotating-frame description,

$$H_{\text{drive}} = \frac{\Omega}{2}\sum_j \left(|g_j\rangle\langle r_j| + |r_j\rangle\langle g_j|\right) - \Delta\sum_j n_j,$$

where

$$n_j = |r_j\rangle\langle r_j|.$$

For two atoms, the interaction contribution is

$$H_{\text{int}} = V(R)n_1 n_2.$$

Thus the idealized two-atom Hamiltonian can be summarized as

$$H(R) = \frac{\Omega}{2}\sum_{j=1}^{2} \left(|g_j\rangle\langle r_j| + |r_j\rangle\langle g_j|\right) - \Delta\sum_{j=1}^{2} n_j + V(R)n_1 n_2.$$

When $V(R) \gg \Omega$, the $|rr\rangle$ state is shifted away from resonance and double excitation is suppressed. However, for finite $V(R)$, the double-excitation amplitude is not identically zero. In the strong-blockade limit, the leakage scale is often qualitatively associated with an off-resonant factor of order $(\Omega/V)^2$, with the actual final-shot probability depending on the pulse shape and measurement time.

Bloqade Analog Workflow

Program Construction

For atom spacing R, the two-site geometry is

$$(0,0), \quad (R,0).$$

The Bloqade Analog builder chain is

$$\texttt{start} \to \texttt{add\_position} \to \texttt{rydberg} \to \texttt{rabi.amplitude.uniform.constant} \to \texttt{detuning.uniform.constant}.$$

```
from bloqade.analog import start

program = (
start
.add_position([(0.0, 0.0), (R_um, 0.0)])
.rydberg
.rabi.amplitude.uniform.constant(rabi_amplitude, duration)
.detuning.uniform.constant(detuning, duration)
)

batch = program.bloqade.python().run(shots=shots)
report = batch.report()
```

The notebook uses `report.counts()` to extract final measurement statistics from the local Bloqade Analog emulator.

Measurement Convention

The local Bloqade post-sequence convention used in the notebook is

$$1 \to g, \qquad 0 \to r.$$

For two atoms,

$$\begin{aligned} 11 &\rightarrow gg, \\ 10 &\rightarrow gr, \\ 01 &\rightarrow rg, \\ 00 &\rightarrow rr. \end{aligned}$$

From sampled counts N_{ab}, the empirical probabilities are

$$P(ab) = \frac{N_{ab}}{N_{\text{shots}}}, \qquad ab \in \{gg, gr, rg, rr\}.$$

Because the notebook uses finite-shot sampling, observing zero $|rr\rangle$ counts means that the event was not observed in the sampled shots. It should be interpreted as

$$P_{rr} \text{ below the current sampling resolution,}$$

not as a mathematical proof that $P_{rr} = 0$. Increasing N_{shots} or extending the spacing range can reveal rare double-excitation events.

Performance-Relevant Metrics

The central blockade indicator is the double-excitation probability

$$P(rr).$$

A simple suppression metric is

$$S_{\text{blockade}} = 1 - P(rr).$$

The notebook also tracks the single-excitation probability

$$P_{\text{single}} = P(gr) + P(rg).$$

For a well-timed blockade pulse, one expects large P_{single}, suppressed $P(rr)$, and possibly small $P(gg)$, because population has been transferred out of the initial ground state into the collective single-excitation manifold.

The spacing sweep produces a compact dataset:

$$\{R,\ V(R)/2\pi,\ P(gg),\ P(gr),\ P(rg),\ P(rr),\ P_{\text{single}},\ 1 - P(rr)\}.$$

This is not yet a complete system-level benchmark, but it is a useful minimal observable because it connects a hardware-level variable, atom spacing, to a measurable probability-level outcome.

Notebook Design

The simulator is organized into three cells.

Cell 1: Tools

The first cell contains reusable functions:

- physical constants, including C_6;
- Bloqade program construction;
- Rydberg interaction-scale calculation;
- angular-frequency to MHz conversion;
- local Bloqade execution;
- report-count extraction;
- bitstring normalization;
- probability mapping;
- spacing sweep;
- nearest-spacing selection;
- compact two-column summary printing.

Cell 2: Interactive User Controls

The second cell defines an `ipywidgets` control panel for the user-facing sweep and sampling parameters:

$$R_{\min}, \quad R_{\max}, \quad N_R, \quad R_{\text{selected}}, \quad N_{\text{shots}}.$$

The physics parameters remain fixed in the current user-facing version:

$$\Omega = 2.0, \qquad \Delta = 0.0, \qquad T = 1.0.$$

This keeps the notebook focused on the blockade-performance question rather than turning it into a general-purpose Rydberg simulator.

Cell 3: Run Button, Graphics, and Summary

The third cell defines a run button. Pressing the button executes the spacing sweep and generates:

1. raw Bloqade counts at the selected spacing;
2. mapped probabilities at the selected spacing;
3. $P(rr)$ versus atom spacing R;
4. $V(R)/2\pi$ in MHz versus atom spacing R;
5. final probability distribution at the selected spacing;
6. a compact two-column numerical summary;
7. the resulting dataframe.

The selected spacing is marked with a vertical guide line on the sweep plots. This connects the global spacing dependence to the local probability distribution used as the performance point.

> **Validation role of the UI.** Extending $R_{\max}$ provides a simple sanity check. If the simulator is restricted to the strong-blockade regime, P_{rr} may appear to be zero across the sweep. When the spacing range is extended far enough, P_{rr} becomes visibly nonzero. This shows that suppressed P_{rr} is a physical operating-window result, not a hard-coded or structural zero.

Interpretation

The essential interpretation is that atom spacing controls the interaction scale:

$$V(R) \propto R^{-6}.$$

Small changes in R can therefore produce large changes in $V(R)$. When the interaction scale is large compared with the Rabi drive, the doubly excited state is energetically shifted and double excitation is suppressed. This appears as reduced $P(rr)$. As R increases, $V(R)$ falls rapidly; the blockade weakens, and $P(rr)$ can increase.

> **Performance-modeling interpretation.**
>
> geometry parameter $\rightarrow$ interaction scale $\rightarrow$ sampled probability distribution
>
> $\rightarrow$ blockade leakage $\rightarrow$ architecture-relevant metric.

The selected performance point should be read as a final-time measurement result. The pulse duration matters: for the right timing, population initially in $|gg\rangle$ can be transferred mostly into the single-excitation manifold while $|rr\rangle$ remains suppressed. Thus a result with large $P(gr) + P(rg)$, small $P(gg)$, and low $P(rr)$ is not paradoxical. It indicates that the pulse ends near a collective single-excitation point under blockade.

The broader architectural lesson is that neutral-atom quantum control rewards geometries in which the desired Rydberg interaction is strong on the timescale of the applied pulse. Shorter distances increase $V(R)$, strengthening blockade and supporting compact hardware layouts. However, because the interaction scales as R^{-6}, the useful design region is a crossover rather than a binary constraint. Device design must balance compactness with addressability, crosstalk, optical access, rearrangement protocols, and the intended algorithmic graph.

This is the same modeling pattern needed in larger system-level studies, where device parameters and architecture constraints must be mapped to workload-level quantities such as success probability, fidelity, throughput, or time-to-solution.

Limitations and Extensions

The current version is intentionally minimal. It does not yet include atom loss, finite temperature, calibration drift, laser noise, readout error, waveform distortion, hardware scheduling overhead, or direct comparison with experimental data. It also reports final-shot probabilities after a fixed pulse, not full continuous-time population dynamics.

Natural extensions include:

1. finite-shot uncertainty bands for $P(rr)$;
2. spacing disorder $R \to R + \delta R$;
3. imperfect filling or atom-loss models;
4. optional advanced controls for Ω, Δ, and T;
5. pulse-duration dependence;
6. time-resolved population diagnostics;
7. threshold-based success metrics;
8. larger arrays or graph-based constraints;
9. benchmark reports that record model assumptions and run metadata.

A particularly useful next step is a geometry-jitter study,

$$R \sim \mathcal{N}(R_0, \sigma_R),$$

followed by repeated emulation or probability resampling. This would convert the deterministic spacing sweep into a robustness study:

$$\sigma_R \to \text{distribution of } P(rr) \to \text{reliability metric.}$$

Takeaway

This demonstration is a compact Bloqade Analog workflow for neutral-atom performance modeling. It shows how a hardware-level parameter, atom spacing, determines a physical interaction scale, how that scale influences blockade behavior, and how the result can be summarized through probability-level metrics such as $P(rr)$, P_{single}, and $1 - P(rr)$.

> **Modeling chain.**
>
> atom geometry $\to$ Rydberg interaction $\to$ analog quantum dynamics
> $\to$ measurement probabilities $\to$ performance-relevant metric.

The value of the demo is not its complexity. The value is the modeling discipline: define the minimal useful model, expose assumptions, compute interpretable metrics, add interactive validation checks, and connect device physics to algorithmic behavior. This workflow is directly extensible to more realistic neutral-atom studies involving noise, disorder, calibration uncertainty, larger geometries, and workload-level benchmarking.

References

1. QuEra Computing, *Bloqade Documentation*. `https://bloqade.quera.com/`
2. QuEra Computing, *Bloqade Analog Documentation*. `https://queracomputing.github.io/bloqade-analog/`
3. QuEra Computing, *Rydberg Blockade Tutorial*. `https://queracomputing.github.io/Bloqade.jl/dev/tutorials/1.blockade/main/`